Reverse Speed

To obtain the reverse gear, the synchronizer S_1 is actuated to move right towards the reverse gear 6 to engage as shown in Figure 3.5.3.5. As the idler gear is placed between the gears 5 in the lay shaft and 6 in the main shaft the idler gear changes the direction of rotation of the gear 6 in the main shaft. Now the vehicle moves in the reverse direction. The power transmitted from the clutch shaft to the main shaft is through gears 1-2-5-Idler-6-S_1.

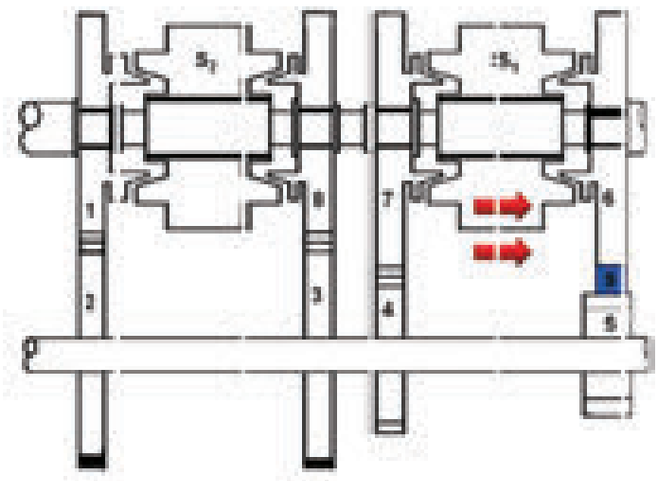


Figure 3.5.3.5 Reversed gear condition

3.5.4 Synchronizing unit

In synchronizing unit two cone shaped parts are present. One cone is a part of the collar and the second cone is a part of the gear. Both the cones rotate at different speeds. An exploded view of the synchronizer unit is shown in Figure.

When the cone 2 is rotated, cone 1 start to rotate and move towards the cone 2 and be in contact with the cone 2. Because of this action friction occurs in the contact area of the cones and hence it increases the speed of gear. Now the both cone speeds come to the same. Further moving the collar makes the outer drum to contact with the dog clutch in the gear. Now the rotational power is transmitted to the main shaft. By the same way it moves on left hand side and contacts with the another gear and transmits the power to the main shaft. The cone 2 is locked under different states by the ball and spring in the cone 1. It is connected to the gear changing fork in the slot which is located in the outer sliding sleeve.

Uses of synchromesh unit

1. For operating the gears smoothly by synchronizing two different speeds of the gears.
2. For changing gears easily
3. No need of double de clutching
4. For transmitting power efficiently
5. Gears will not wear rapidly
6. Less noise in operation

De merits of synchromesh unit

1. Low cost
2. Maintenance is difficult
3. Difficult repair

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